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Q.1) Consider a software project with Five tasks T1 - T5. Duration of the Five tasks in weeks 3, 2, 3, 5 & 2 respectively. T2 & T4 can start when T1 is complete. T3 can start when T2 is complete. T5 can start when both T3 & T4 are complete. Draw the cpm network representation of the project. When is the latest start due date of the Task 3? What is the float time of the Task 4? Which tasks are on the critical path?

Solutions:

Task	Duration
T1	3
T2	2
T3	3
T4	5
T5	2

Given,
T2 & T4 can start when T1 is complete & T3 can start when T2 is complete. T5 can start when T3 or T4 is complete

CPM Network Diagram:

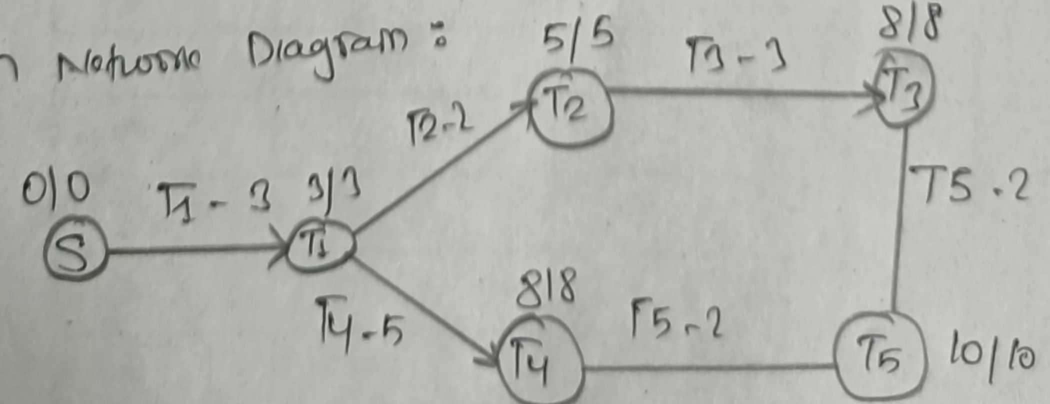


Table:

Task	Duration	ES	Ef	LS	Lf	Float
T_1	3	0	3	0	3	0
T_2	2	3	5	3	5	0
T_3	3	5	8	5	8	0
T_4	5	3	8	3	8	0
T_5	2	8	10	8	10	6

Critical path:

Task with float=0

$\therefore$ $T_1 \to T_2 \to T_3 \to T_5$ & $T_1 \to T_4 \to T_5$ are the critical paths

The latest start date of T_3 is 5 weeks

Q2) From the following data using PERT:

a) Determine critical path of the project.

b) Calculate variance & standard deviation of all activities

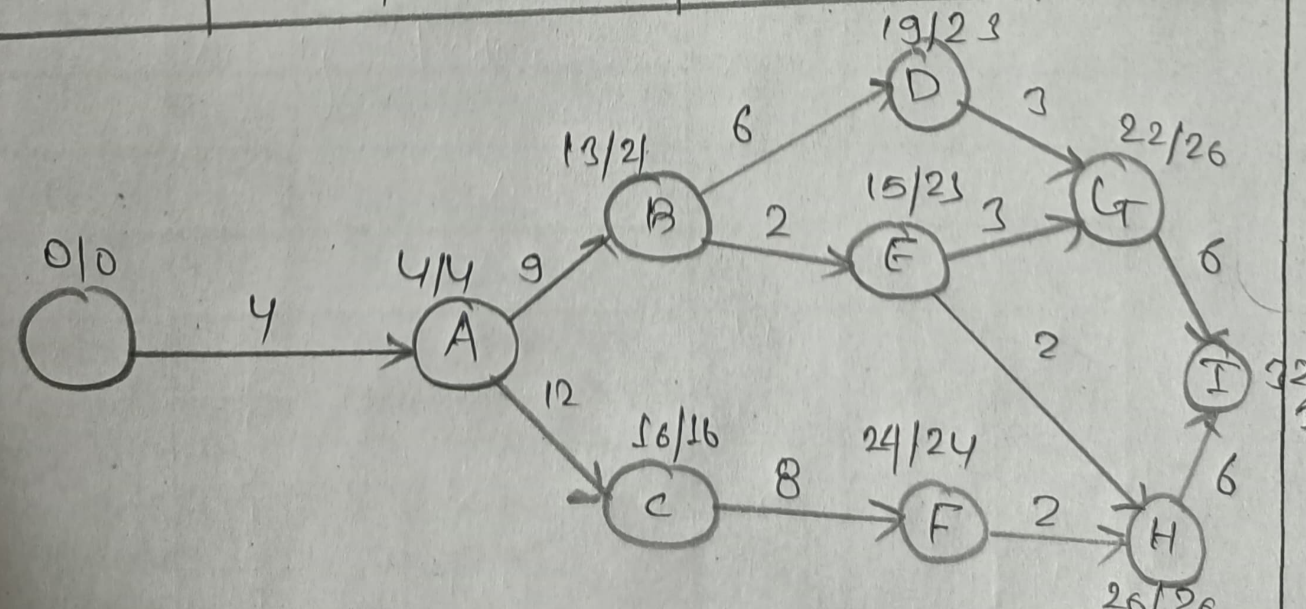
c) Calculate the probability of completing the project in 16 days (download standard normal distribution table & match the corresponding value of z to that probability value for the answer.

Activity	Predecessor	Timing		
		A	M	B
A		1	5	3
B	A	2	8	20
C	A	8	12	16
D	B	4	6	8
E	B	1	2	3
F	C	6	8	10
G	E, D	2	3	4

H	E, D	2	3	4
I	E, F	2	2	2

Solution :

Activity	te	variance (σ^2)	Std. Deviation
A	1+20+ 3/8 = 4	0.111	0.33
B	2+ 32+ $\frac{20}{6}$ = 9	9	3
C	8+48+ 10/6 = 12	1.778	1.3
D	4+24+ 8/6 = 6	0.444	0.667
E	1+8+ 3/6 = 2	0.111	0.333
F	6+ 32+10/6 = 8	0.444	0.667
G	2+ 12 + 4/6 = 3	0.111	0.33
H	2+ 8+ 2/6 = 2	0	0
I	6 + 24+ 6/6 = 6	0	0



a) Critical path: $A \to C \to F \to H \to I$
Duration : 32 days

b) Probability of completion in 36 days
Variance of path = $A \to C \to F \to H \to I$
Sum = $0.111 + 1.778 + 0.444 + 0 + 0$
$= 2.333$

Computing Z value, $Z = \frac{36 - 32}{1.528} = 2.62$

Now,
Z value in table for $Z_{2.62} = 99.55\%$.
Probability.

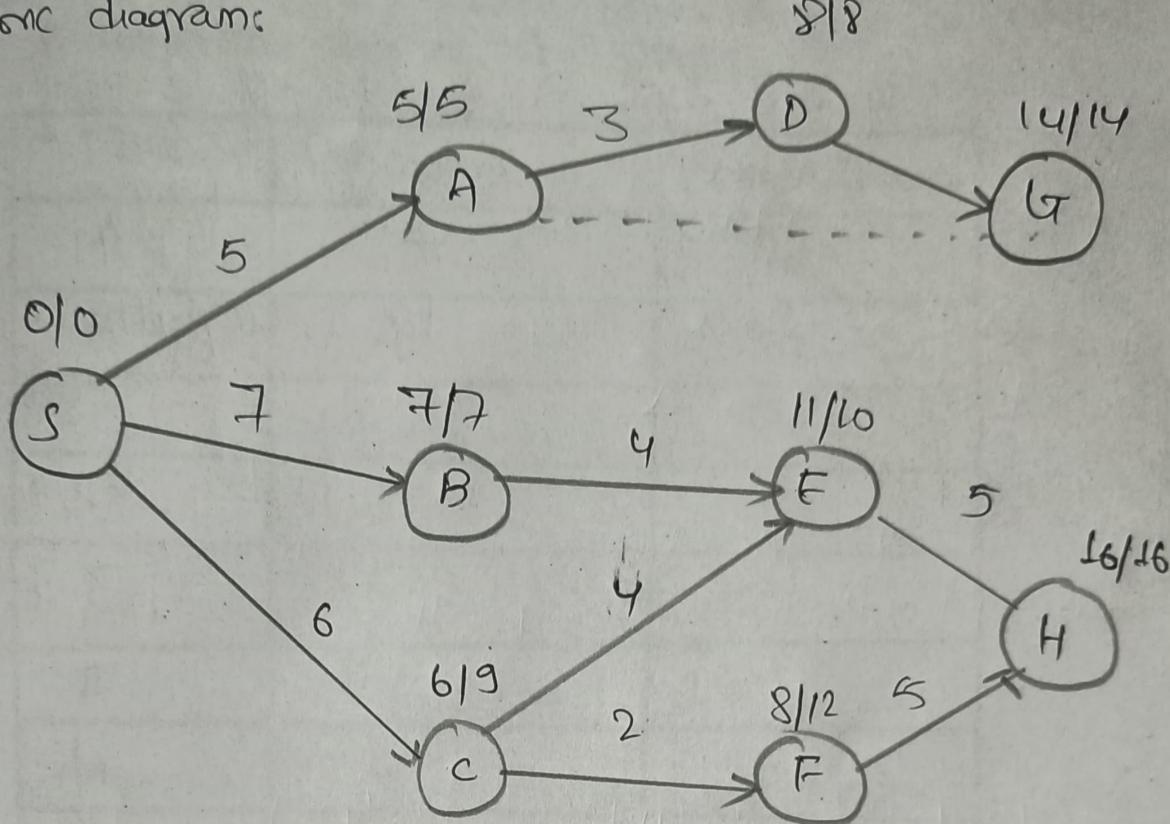
Q3) A project consist of 8 activities. The activity completion times and the precedence relationships are as follows:

Activity	Completion time (days)	Immediate predecessor activities
A	5	—
B	7	—
C	6	—
D	3	A
E	4	B, C
F	2	C
G	6	A, D
H	5	E, F

a) Draw the network diagram
b) Calculate the minimum overall project completion time & identify which activities are critical.
c) If activity E is delayed by 3 days how is the project completion time affected?
d) If activity F is delayed by 1 days how is the project completion time affected?

Solution:

Network diagram:



- If Activity E is delayed by 3 days the project duration increases by 3 days since E lies in critical path & the float of item in the given critical path is 0 days. Project duration completion = 19 days.
- If activity E is delayed by 3 days project increases by 0 days because.

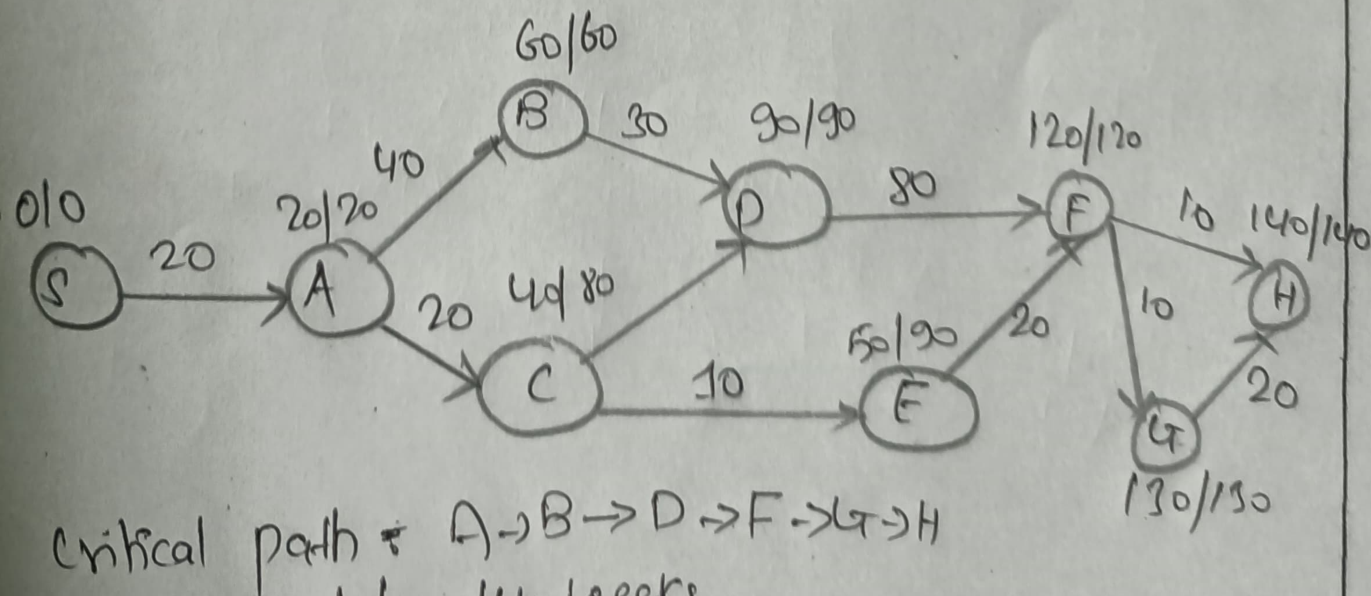
$$Float(F) = LS_F - ES_F = 11-8 = 3$$

Since float of activity is 3 days we have 3 days more for completion. Project completion = 16 days

Q4) Draw a precedence network diagram by performing forward & backward passes. Also determine the critical path & completion date of the project.

Task	Precedents	Duration (weeks)
A	–	20
B	A	40
C	A	20
D	C, B	30
E	C	10
F	D, E	30
G	E	10
H	F, G	10

Solution:



Critical path : A → B → D → F → G → H
completion date = 140 weeks.